

ADAM — MASTER BUSINESS MODEL, DUAL-TERRITORY MARKET ANALYSIS & INTERNATIONAL GTM

Commercialization Blueprint · Actual BOM (₹6,500.35) · India vs International TAM/SAM/SOM · Export Costing

Developer: Dgen Technologies Private Limited (Kolkata, India) | Version 4.0 (Global Dual-Territory Architecture)

Official BOM Reference: https://docs.google.com/spreadsheets/d/10W3NfeC3rfsyMWaw4OGP7p-bOafOCyfFovaU_Fv0rMg/edit?usp=sharing

1. Executive Summary & Dual-Territory Strategic Realignment

ADAM (Autonomous Desktop AI Module) is India’s first domestically engineered and manufactured embodied AI desk companion. This master document establishes the dual-territory commercial model for Domestic India (₹20,000–₹25,000 band, blended ₹22,500 ASP) and International Export markets (\$248–\$349 USD, blended \$320 USD ~₹27,040 INR). Incorporating the audited BOM of ₹6,500.35 from `ADAM BOM.xlsx`, international export yields ₹10,856+ net contribution per unit (+35.8% higher than domestic) due to zero-rated GST export incentives.

Metric	Domestic India Market	International Export Market	Strategic Delta & Rationale
Retail Price Band (Gross)	₹20,999 – ₹24,999	\$248 – \$349 USD (~₹20,950 – ₹29,500)	Standard vs Pro hardware editions
Blended Retail ASP	₹22,500 (incl. 18% GST)	\$320 USD (~₹27,040 INR)	International captures higher USD purchasing power
Net Realized Hardware Revenue	₹19,068 (net of GST)	\$303.50 USD (~₹25,646 INR)	Net of 5% Merchant of Record fee
Actual Hardware BOM COGS	₹6,500.35 (\$76.93)	₹6,500.35 (\$76.93)	Direct audited BOM from ADAM BOM.xlsx
Assembly, QA & Packaging	₹500.00	₹500.00	Factory assembly & 48-hr QA burn-in in Kolkata
Manufacturing Gross Margin (%)	65.91% (Direct) / 62.24% (Loaded)	74.65% (Direct) / 71.90% (Loaded)	Zero-rated export GST expands international margin
Outbound Express Shipping	₹350 (Domestic Air)	₹1,859 (\$22.00 - ShiprocketX Air)	800g package air express from Kolkata CCU hub
Net Contribution Cash / Unit	₹7,995.45 / unit (41.93%)	\$128.48 / unit (₹10,856 / 41.2%)	+₹2,861 (+35.8%) more operating cash per international unit!
3-Year Addressable SOM (Units)	54,050 units	44,720 units	Combined global 3-year production pool: 98,770 units
3-Year Hardware Revenue Pool	₹121.63 Crore	\$14.31 Million USD (~₹120.92 Cr)	Combined Global Ecosystem Revenue: ₹242.55 Crore

2. Actual Production Bill of Materials (BOM) Breakdown (`ADAM BOM.xlsx`)

BOM Section	Description	Component Count	Line Items	Subtotal (₹)	Subtotal (\$ USD)	% of Total
Section A	Electronics & Sensors (Compute, MCU, Audio, Vision, Radar)	105	67	₹5,352.00	\$63.34	82.33%
Section B	Mechanical & Hardware (Shell, Acrylic, Screws, Bearing, Inserts)	36	8	₹783.35	\$9.27	12.05%
Section C	Packaging, Accessories & Documentation (Charger, Cable, Box, Foam)	5	5	₹365.00	\$4.32	5.62%
GRAND TOTAL	Complete Ready-to-Assemble Physical Product	146	74	₹6,500.35	\$76.93	100.00%

3. Domestic Unit Economics & Contribution Waterfall across Price Corridors

Waterfall Line Item	Standard (₹20,999)	Blended Target (₹22,500)	Pro (₹24,999)	Strategic Context
Gross Customer Billing (Invoice)	₹20,999.00	₹22,500.00	₹24,999.00	Customer retail price inclusive of GST
(-) 18% GST (HSN 8471)	-₹3,203.24	-₹3,432.20	-₹3,813.41	Remitted to Government of India
Net Realized Revenue	₹17,795.76	₹19,067.80	₹21,185.59	100.0% Net Top-Line Revenue
(-) Actual Hardware BOM (BOM.xlsx)	-₹6,500.35	-₹6,500.35	-₹6,500.35	Direct BOM (Electronics + Mech + Pkg)
DIRECT GROSS MANUFACTURING PROFIT	₹11,295.41	₹12,567.45	₹14,685.24	Direct manufacturing margin on materials
DIRECT GROSS MARGIN (%)	63.47%	65.91%	69.32%	Industry-leading hardware profitability
(-) SMT Assembly & QA Burn-in	-₹500.00	-₹500.00	-₹500.00	Assembly labor & 48-hr QA testing
(-) Inward Freight & Buffer	-₹200.00	-₹200.00	-₹200.00	Supplier freight to Kolkata facility
LOADED FACTORY GROSS PROFIT	₹10,595.41	₹11,867.45	₹13,985.24	Profit after full assembly loading
LOADED GROSS MARGIN (%)	59.54%	62.24%	66.01%	62.24% Loaded Manufacturing Margin
(-) Outbound Doorstep	-₹350.00	-₹350.00	-₹350.00	Insured air express

Express Courier				delivery
(-) Razorpay Payment Gateway (2%)	-₹420.00	-₹450.00	-₹500.00	Payment processing fee
(-) 1-Year Warranty & RMA (3%)	-₹534.00	-₹572.00	-₹635.00	Component defect & swap reserve
(-) Customer Acquisition Cost (CAC)	-₹2,500.00	-₹2,500.00	-₹2,500.00	Blended marketing & creator seeding
NET CONTRIBUTION MARGIN (PER UNIT)	₹6,791.41	₹7,995.45	₹9,990.24	Operating cash flow retained per unit
NET CONTRIBUTION MARGIN (%)	38.16%	41.93%	47.16%	Cash flow retained after all marketing

4. Market Analysis Part 1: Domestic Indian Market Multi-Segment Funnel

Domestic Segment	Total Universe	AI Filter (%)	AI User Base	Desk Prop. (%)	Addressable SAM	Conv. (%)	3-Yr SOM	ASP (₹)	Revenue (₹ Cr)
1. Vibe Coders (AI Devs)	6.00 M	68.33%	4,100,000	10.00%	410,000	3.00%	12,300	₹22,500	₹ 27.68 Cr
2. Remote Knowledge Workers	15.00 M	92.00%	13,800,000	21.74%	3,000,000	0.67%	20,000	₹22,500	₹ 45.00 Cr
3. Startup Founders / Solos	1.50 M	74.00%	1,110,000	13.51%	150,000	2.00%	3,000	₹22,500	₹ 6.75 Cr
4. STEM / Robotics Higher Ed	6.00 M	72.00%	4,320,000	9.26%	400,000	1.38%	5,500	₹22,500	₹ 12.38 Cr
5. Creators & Studios	2.50 M	86.00%	2,150,000	11.63%	250,000	1.50%	3,750	₹22,500	₹ 8.44 Cr
6. Corporate B2B / GCC Desks	0.05 M	80.00%	40,000	15.00%	6,000	100.0%	6,000	₹22,500	₹ 13.50 Cr
7. Smart Home (MQTT Hub)	3.50 M	48.60%	1,701,000	20.58%	350,000	1.00%	3,500	₹22,500	₹ 7.88 Cr
SUBTOTAL DOMESTIC INDIA	34.55 M	—	27.22 M	—	4.57 M	1.18%	54,050	₹22,500	₹121.63 Cr

5. Market Analysis Part 2: International Global Market Multi-Segment Funnel

Global Segment	Global TAM	AI Filter (%)	AI User Base	Desk Prop. (%)	Addressable SAM	Conv. (%)	3-Yr SOM	ASP (\$ USD)	Revenue (\$ USD)	Revenue (₹ Cr)
1. Global Software Developers	28.70 M	76.00%	21,812,000	5.00%	1,090,600	1.00%	10,900	\$320	\$3,488,000	₹ 29.47 Cr
2. Remote Knowledge Workers	300.00 M	75.00%	225,000,000	2.00%	4,500,000	0.25%	11,250	\$320	\$3,600,000	₹ 30.42 Cr
3. Tech Founders & Solos	150.00 M	74.00%	111,000,000	1.00%	1,110,000	0.50%	5,550	\$320	\$1,776,000	₹ 15.01 Cr
4. STEM Higher Education	65.00 M	72.00%	46,800,000	2.50%	1,170,000	0.50%	5,850	\$320	\$1,872,000	₹ 15.82 Cr
5. Digital Creator Economy	50.00 M	86.00%	43,000,000	3.50%	1,505,000	0.50%	7,525	\$320	\$2,408,000	₹ 20.35 Cr
6. Smart Home Power Users	2.50 M	48.60%	1,215,000	15.00%	182,250	2.00%	3,645	\$320	\$1,166,400	₹ 9.86 Cr
TOTAL INTERNATIONAL	596.20 M	—	448.83 M	—	9,557,850	0.47%	44,720	\$320	\$14,310,400	₹120.92 Cr

6. Comparative Sizing: Domestic India vs. International Markets

Dimension / Parameter	Domestic Indian Market	International Export Market	Combined Global Potential
Total Population Universe (TAM)	34.55 Million	596.20 Million	630.75 Million total universe
Active AI Agent User Pool	27.22 Million (78.8% blended)	448.83 Million (75.3% blended)	476.05 Million AI adopters
Addressable Desk Tech SAM	4.57 Million desks	9.56 Million desks	14.13 Million addressable desks
3-Year Penetration SOM (Units)	54,050 Units	44,720 Units	98,770 Units global production
Blended Retail ASP	₹22,500 INR	\$320 USD (~₹27,040 INR)	—
Gross 3-Year Hardware Revenue	₹121.63 Crore	\$14.31 Million USD (~₹120.92 Cr)	₹242.55 Crore (\$28.7M USD)
Retained Operating Cash Margin	₹43.21 Crore (35.5%)	₹49.80 Crore (\$5.89M USD / 41.2%)	₹93.01 Crore (\$11.0M USD)

7. International Go-To-Market (GTM) Strategy & Export Costing Architecture

1. Export GST & Statutory Clearances: Under Section 16 of the IGST Act, exports clear customs at 0% IGST under Form GST RFD-11 (LUT). Unutilized input tax credit of ₹850–₹1,100 per unit is refunded in cash under Rule 89(4). BIS CRS is 100% exempt for export goods per DGFT/MeitY.
2. International Freight Economics: An 800g package (200×150×100mm) billed at 1.0kg costs \$17–\$22 (₹1,450–₹1,850) via ShiprocketX B2C Air to the USA, or \$21–\$29 via India Post Speed Post EMS. DHL Express / FedEx Priority ranges from \$31–\$43.

3. Merchant of Record (MoR): Lemon Squeezy / Paddle (5% + \$0.50) acts as legal reseller, handling US sales tax, EU VAT, and UK VAT automatically.

4. Customs Duties & De Minimis: US Section 321 grants 100% exemption for shipments under \$800 USD (0% duty, 0% tax). EU/UK computers under HSN 8471 have 0% customs duty under WTO ITA.

5. Compliance & Lab Testing: FCC Part 15B (modular approval using pre-certified RPi Zero 2W and ESP32-S3) + CE RED + RoHS 3 is tested locally at SAMEER Kolkata (Salt Lake Sector V) for ₹3.5L–₹4.8L (\$4.2k–\$5.8k USD) in 4–5 weeks.

6. Crowdfunding Comps: Loona Petbot (\$3.0M, 7.8k backers), Loona Deskmate (\$721k, 3.1k backers), Eilik (\$1.08M, 14k orders), Flipper Zero (\$4.88M, 37.9k backers). BackerKit pledge manager expands post-campaign revenue by +18%.

Export Waterfall Line Item	Early Bird (\$248 / ₹20,956)	Standard (\$320 / ₹27,040)	Strategic Rationale / Basis
Gross Customer Retail Invoice	\$248.00	\$320.00	FOB DTC Export Retail Price (excl. VAT)
(-) Merchant of Record Fee (5% + \$0.50)	-\$12.90	-\$16.50	Lemon Squeezy / Paddle (global tax filing)
Net Realized Export Revenue	\$235.10 (₹19,866)	\$303.50 (₹25,646)	Remitted to Indian corporate bank account
(-) Actual Hardware BOM (BOM.xlsx)	-\$76.93 (₹6,500.35)	-\$76.93 (₹6,500.35)	Audited production BOM (Electronics+Mech+Pkg)
(-) SMT Assembly & 48-hr QA Labor	-\$5.92 (₹500.00)	-\$5.92 (₹500.00)	Factory labor in Kolkata assembly plant
(-) Domestic Inward Freight	-\$2.37 (₹200.00)	-\$2.37 (₹200.00)	Inward component logistics to facility
LOADED MANUFACTURING PROFIT	\$149.88 (₹12,665)	\$218.28 (₹18,445)	63.7% – 71.9% Loaded Manufacturing Margin
(-) Outbound Air Express Courier	-\$22.00 (₹1,859)	-\$22.00 (₹1,859)	ShiprocketX Air from Kolkata CCU to US/EU
(-) Destination Customs Duty	\$0.00	\$0.00	US Section 321 (\$800 de minimis) / EU WTO ITA 0%
(-) 1-Year Global Warranty & RMA (4%)	-\$9.92 (₹838)	-\$12.80 (₹1,082)	Component replacement reserve
(-) International CAC (Kickstarter/Ads)	-\$40.00 (₹3,380)	-\$55.00 (₹4,648)	Global video PR, digital ads & seeding
NET EXPORT CONTRIBUTION (PER UNIT)	\$77.96 / unit	\$128.48 / unit	Retained operating profit per unit
NET EXPORT CONTRIBUTION IN INR	₹6,587.62 / unit	₹10,856.56 / unit	+₹2,861 (+35.8%) more cash than domestic!

8. Google Cloud Vertex AI Costing & AI Credits Economics

Modality / Operation	Vertex Rate (USD / 1M)	Rate in INR (per 1M)	Active Minute Cost (₹)
Live Audio Input (Speech)	\$3.00 / 1M tokens	₹253.50 / 1M	₹0.211 / min (~30s speech)
Native Audio Output (Speech)	\$12.00 / 1M tokens	₹1,014.00 / 1M	₹0.761 / min (~30s speech)
Attention-Gated Vision (0.125 FPS)	\$1.00 / 1M tokens	₹84.50 / 1M	₹0.163 / min (1 frame / 8s)
Prompt Context Caching	\$0.0375 / 1M tokens	₹3.17 / 1M	₹0.150 / min (95% prompt discount)
Function Calling / Tools	\$3.00 / 1M tokens	₹253.50 / 1M	₹0.180 / min (Needle 3)
Hybrid Search Pipeline	Free DDG + \$14/1k Grounding	—	₹0.080 / min (90% DDG / 10% Grounding)
Cloud Run Token Broker & Egress	Fastify asia-south1	—	₹0.250 / min (Mumbai deployment)
Payment Gateway Fee (2%)	Razorpay UPI / Cards	—	₹0.055 / min
Total Loaded Operational Cost	—	—	₹1.850 / active minute

9. Global Competitive Landscape (Desktop Robots)

Competitor Robot	Manufacturer	MSRP (USD)	Landed INR (incl. GST)	Subscription Model	ADAM Competitive Advantage
Loona Petbot	KEYi Tech	\$499 – \$529	₹49,500 – ₹52,500	Basic cloud included	Pet toy only; no coding agent hooks, closed firmware.
Eilik / DQ	EnergizeLab	\$139 – \$209	₹13,800 – ₹20,700	No Cloud AI / No LLM	Mechanical desk toy; zero speech/vision intelligence.
Vector 2.0	Digital Dream Labs	\$299 – \$349	₹29,600 – ₹34,500	\$11.99–\$14.99/mo mandatory	Bricks without subscription; high recurring churn.
ADAM Standard	Dgen Technologies	\$248 (₹20,999)	₹20,999 (incl. GST)	No Subscription Required	Agentic IDE hooks, local MQTT, non-bricking Lite mode.
ADAM Pro	Dgen Technologies	\$295 (₹24,999)	₹24,999 (incl. GST)	No Subscription Required	Raspberry Pi CM4, 600 min bundled credits, dual servos.

10. Sourced References & Legal / Statutory Citations

1. Hardware & Packaging Costing: ADAM BOM.xlsx / Google Sheets Master BOM (https://docs.google.com/spreadsheets/d/1OW3NfeC3rfsyMWaw40GP7p-bOafOCyfFovaU_Fv0rMg/edit?usp=sharing) Total BOM: ₹6,500.35, Section A Electronics: ₹5,352.00, Section B Mechanical: ₹783.35, Section C Packaging: ₹365.00.

2. Indian Export Taxation & GST Statutes: Section 16 of Integrated Goods and Services Tax (IGST) Act,

2017; Rule 96A and Rule 89(4) of Central Goods and Services Tax (CGST) Rules, 2017 (LUT Form GST RFD-11 & ITC Refund Form GST RFD-01).

3. Statutory Export Clearances & Exemptions: DGFT Foreign Trade Policy (FTP) 2023; MeitY Notification No. 13/2012-IT & Circular No. 01 of 2014 (100% Export Exemption from BIS Compulsory Registration Scheme); WPC ETA-SD Self-Declaration Guidelines.

4. United States Customs Statutes: 19 U.S.C. § 1321 (Section 321 of the Tariff Act of 1930 - \$800 USD De Minimis Exemption); US Customs and Border Protection (CBP) Entry Type 86.

5. European Union & UK Import Regulations: WTO Information Technology Agreement (ITA-1) 0% Duty on HSN 8471; EU Council Directive 2006/112/EC (VAT Directive & IOSS); UK Global Tariff (UKGT) 2025/2026.

6. Hardware Testing & Compliance Standards: Title 47 CFR Part 15 Subpart B (FCC SDoC); Radio Equipment Directive (RED) 2014/53/EU (EN IEC 62368-1, ETSI EN 301 489, ETSI EN 300 328); RoHS 3 Directive 2011/65/EU / (EU) 2015/863 (EN IEC 63000:2018). Laboratory testing benchmarks from SAMEER Kolkata (MeitY Lab, Salt Lake Sector V).

7. Developer Population Demographics: GitHub Octoverse Annual Report (2024–2026, 150M+ global developers, 21.9M–27.0M in India); Evans Data Corporation Global Developer Population Survey; Stack Overflow Developer Survey (65,000+ respondents, 76%–84% AI adoption).

8. Workplace & Enterprise Knowledge Worker Studies: Microsoft & LinkedIn Work Trend Index (31,000 respondents, 92% Indian knowledge workers, 75% global); Gartner Digital Workplace Forecast; NASSCOM GCC Strategic Report (2,100+ GCCs in India).

9. Creator Economy & Higher Education Censuses: Goldman Sachs Global Investment Research ('The Creator Economy' \$250B to \$480B); Adobe MAX Creators Survey (86% GenAI usage); AICTE Approval Process Handbook 2025–2026 (15.98 Lakh approved seats); UNESCO Institute for Statistics (UIS).

10. Crowdfunding Benchmark Records: Kickstarter Project Archives for Loona Petbot (\$3.0M, 7,811 backers), Loona Deskmate (\$721k, 3,166 backers), Eilik (\$1.08M, 14,000+ orders), Peto Bittle (\$567k, 2,052 backers), Flipper Zero (\$4.88M, 37,987 backers).

11. Google Cloud Vertex AI Live API Pricing: Audio in \$3/1M, Audio out \$12/1M, Video in \$1/1M, Context caching \$0.0375/1M, Search \$14/1k (Google Cloud Official Pricing).